Ji Jade Qiu

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RESEARCH INTERESTS

Reliable Machine Learning, LLM Evaluation, and Generalization

EDUCATION

Columbia University Aug 2025 – Dec 2026

Master of Science in Applied Analytics GPA: 3.70

Relevant Courses: Applied Machine Learning II (A+), Applied Data Science (A), Mathematics for AI (A)

Xi'an Jiaotong-Liverpool University Sep 2021 – May 2025

Bachelor of Science in Financial Mathematics

*Relevant Courses:*Stochastic Processes And Calculus (A), Statistical Distribution Theory (A-)

PUBLICATIONS

- **Ji Qiu**, Yamini Chitikela, Zhenquan Jia, and Chunjiang Zhu. **"Automated Continuous LLM-driven Literature Mining for a Large-Scale Hepatotoxicity Dataset."** In *IEEE International Conference on Bioinformatics & Bioengineering (BIBE) 2026*. Accepted. [code]
- Yamini Chitikela, **Ji Qiu**, and Chunjiang Zhu. **"Generalization of Hypergraph Neural Networks via VC Dimension and Rademacher Complexity."** In *Learning on Graphs Conference (LoG) 2026*, Proceedings Track. Accepted (Poster). [code]

RESEARCH EXPERIENCE

Department of Computer Science, Old Dominion University May 2026 – Aug 2026

Summer Research Intern / Advised by Prof. Chunjiang Zhu

- **Automated LLM-based Literature Mining for a Large-Scale Hepatotoxicity Dataset**

  Built an end-to-end pipeline for automatically mining hepatotoxicity findings from the biomedical literature, spanning literature retrieval, structured extraction, and downstream validation. Collected **53.6k** PubMed/PMC papers through fault-tolerant downloading, deduplication, and incremental ingestion. Designed an LLM extraction workflow with a controlled 14-field schema and anti-hallucination constraints to extract **189k** structured findings across 17k chemicals. Validated the resulting dataset through per-marker QSAR modeling, comparing Graphormer, GraphGPS, and GraphSAGE against RandomForest and XGBoost, with manual review confirming **95.4% core-triple accuracy**. Led the resulting first-author paper at **BIBE 2026**.

- **Generalization of Hypergraph Neural Networks via VC Dimension and Rademacher Complexity**

  Characterized the generalization ability of hypergraph neural networks (HNNs) by connecting their VC dimension to a two-stage hypergraph Weisfeiler–Leman (HWL) coloring framework. Established theoretical bounds relating VC dimension to HWL color complexity, model parameter count, and weight precision, alongside Rademacher complexity bounds characterizing data-dependent generalization. Designed and conducted the full empirical evaluation, including systematic width/depth sweeps and shattering experiments across 10 real and synthetic hypergraph datasets, validating the predicted theoretical trends. Co-authored the **accepted LoG 2026 paper**.

Department of Financial Mathematics, XJTLU Dec 2024 – Apr 2025

Undergraduate thesis / Advised by Prof. Xiangqian Sun

- **LSTM-based Equity Portfolio Optimization with Shannon Entropy Regularization**

  Developed an end-to-end framework integrating deep learning-based return forecasting with portfolio optimization using historical price data from 35 large-cap S&P 500 stocks. Trained stock-specific two-layer LSTM models on 60-day lookback windows, incorporating rolling-window training and an Autoregressive Moving Pointer Model (AMPM) module to capture evolving market dynamics. Selected the top 10 stocks based on predicted returns and applied entropy-regularized portfolio optimization to balance returns and diversification. Out-of-sample evaluation against return-only optimization and random equal-weight baselines demonstrated the **highest returns and lowest maximum drawdown** while maintaining comparable volatility over a 30-day holdout window.

Pacific Insurance Company May 2024 – Aug 2024

Economic Scenario Generator Project

- **Economic Scenario Generator for Insurance Risk Management**

  Developed a Python-based economic scenario generator to simulate stochastic paths of key market variables for insurance asset–liability and solvency risk analysis. Calibrated a nonlinear state-space model using an Unscented Kalman Filter (UKF), enabling stable parameter estimation in regimes where linear Kalman and Extended Kalman Filter (EKF) approaches degrade. Fitted a Cox–Ingersoll–Ross (CIR) model to historical interest-rate data and modeled equity dynamics using geometric Brownian motion (GBM). Evaluated the interest-rate model on the held-out split, achieving a short-rate **15-basis-point RMSE**.

COMMUNITY SERVICE

- Outstanding Class Leader 2021 - 2023
- Student Representative, Department of Financial Mathematics 2022 - 2023
- President, Student Career Development Association, Information Division 2021 - 2022